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Dear Professor García-López, Dr. Ontañon-García,

We are writing to you concerning the below referenced manuscript which you submitted to the Photonics. Based on the completed set of reviews, the manuscript has been accepted in its present format.

Manuscript ID: photonics-3866364

Type of manuscript: Article

Title: Real-time Observer and Neuronal Identification of an Erbium-Doped Fiber Laser

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Lasers, Light Sources and Sensors

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Lasers and Complex System Dynamics

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Science Citation Index Expanded (SCIE)	OPTICS	
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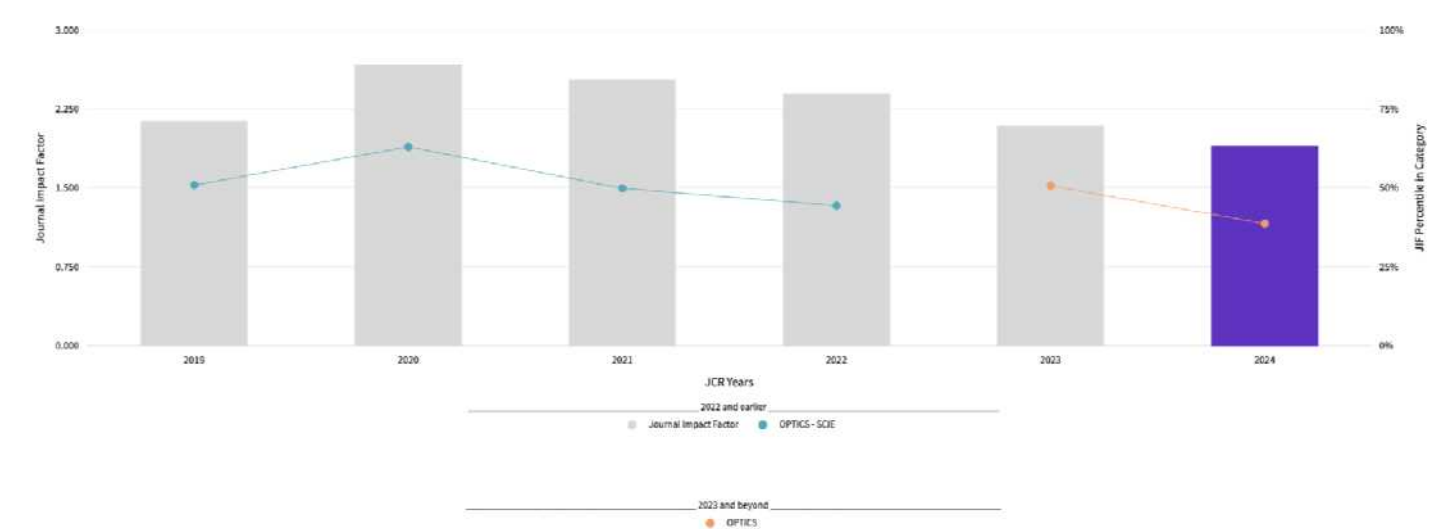
2024 JOURNAL IMPACT FACTOR

1.9

2024 JOURNAL IMPACT FACTOR WITHOUT SELF CITATIONS

1.7

Journal Impact Factor Trend 2024



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CATEGORY

OPTICS

77/125

JCR YEAR	JIF RANK	QUART ILE	JIF PERCENTILE	
2024	77/125	Q3	38.8	
2023	59/119	Q2	50.8	

Rank by JIF before 2023 for OPTICS

EDITION

Science Citation Index Expanded (SCIE)

JCR YEAR	JIF RANK	QUART ILE	JIF PERCENTILE	
2022	56/100	Q3	44.5	
2021	51/101	Q3	50.00	
2020	37/99	Q2	63.13	
2019	48/97	Q2	51.03	

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CATEGORY

OPTICS

64/125

JCR YEAR	JCI RANK	QUART ILE	JCI PERCENTILE	
2024	64/125	Q3	49.20	
2023	56/120	Q2	53.75	
2022	55/119	Q2	54.20	
2021	54/118	Q2	54.66	
2020	54/115	Q2	53.48	
2019	53/114	Q2	53.95	
2018	51/108	Q2	53.24	
2017	54/106	Q3	49.53	

Article

Real-Time Observer and Neuronal Identification of an Erbium-Doped Fiber Laser

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Abstract

This paper presents the implementation of a real-time nonlinear state observer applied to an erbium-doped fiber laser system. The observer is designed to estimate population inversion, a state variable that cannot be measured directly due to the physical limitations of measurement devices. Taking advantage of the fact that the laser intensity can be measured in real time, an observer was developed to reconstruct the dynamics of population inversion from this measurable variable. To validate and strengthen the estimate obtained by the observer, a Recurrent Wavelet First-Order Neural Network (RWFONN) was implemented and trained to identify both state variables: the laser intensity and the population inversion. This network efficiently captures the system's nonlinear dynamic properties and complements the observer's performance. Two metrics were applied to evaluate the accuracy and reliability of the results: the Euclidean distance and the mean square error (MSE), both of which confirm the consistency between the estimated and expected values. The ultimate goal of this research is to develop a neural control architecture that combines the estimation capabilities of state observers with the generalization and modeling power of artificial neural networks. This hybrid approach opens up the possibility of developing more robust and adaptive control systems for highly dynamic, complex laser systems.

Keywords: state observer; erbium-doped fiber laser; recurrent wavelet first-order neural network



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1. Introduction

Erbium-doped fiber lasers (EDFLs) have many applications in optical communications and photonics. These lasers can be compact and tunable. One design incorporates a microfiber coupled to a silicon dioxide microsphere to tune and filter the wavelength [1]. EDFLs are particularly useful for long-distance communication as they amplify optical signals without conversion to electrical signals [2]. Recent research has focused on generating ultrashort pulse widths and high repetition rates. A tunable EDFL based on distributed random feedback has demonstrated a wide tuning range of 40 nm with a narrow linewidth [3]. A novel

configuration called the erbium-doped fiber laser amplifier (EDFLA) combines laser and amplifier functionalities and serves as an optical amplifier, laser source and optical ON/OFF tool [4]. These advances in EDFL technology contribute to improved performance and versatility in optical communication systems and other photonic applications.

The following references include contributions from the designer of the controller for this type of laser [5–7]. In addition, this collection of papers explores various aspects of fiber laser systems and their control. Ref. [8] demonstrated an erbium-doped fiber laser with an option to switch on the operating state, and investigated pulse splitting and multilevel modulation. Ref. [9] focused on process monitoring in selective laser melting and developed a variable retrofocus system and real-time signal acquisition for improved manufacturing processes. Ref. [10] compared proportional–integral–derivative (PID) control with observer-based state feedback (OBSF) for precise positioning of the laser beam, and found OBSF to be superior in disturbance rejection. Ref. [11] presented a new approach that uses the principal polarization states to describe the tunability of fiber lasers, and validated this method for a 4 m fiber laser over a wavelength range of 2 nm. These studies contribute to the overall advancement of fiber laser technology and address challenges in beam control, process monitoring, and system tunability in various applications.

Recurrent Wavelet First-Order Neural Networks (RWFONNs) have proven to be powerful tools for real-time identification and control of complex systems. These networks use a single-layer structure with a single neuron and a Morlet wavelet activation function [12,13]. RWFONNs have proven successful in identifying and controlling robotic manipulators [12] and chaotic systems implemented in Field-Programmable Analog Arrays [13]. In particular, an RWFONN with fixed parameters is able to identify multiple chaotic systems without recalibration [14]. These networks offer advantages in online training and real-time applications without the need for multiple epochs or data batches [13]. In addition, first-order recurrent neural networks have been hierarchically classified based on their expressive power, providing insights into their computational capabilities and attractive behavior [15].

The integration of artificial neural networks with Luenberger observers has been explored for state estimation in nonlinear and uncertain systems. Recurrent neural networks (RNNs) have been used to develop Luenberger-like observers for discrete-time nonlinear systems with partial information [16]. Extended Luenberger observers have been combined with Grossberg’s additive model for dynamic neural networks to provide state estimates for systems with partially known or unknown dynamics [17]. A multilayer recurrent neural network was developed for real-time synthesis of asymptotic state observers in linear time-varying systems [18]. In addition, robust asymptotic neuro-observers with time-delay terms have been developed for unknown nonlinear systems using dynamic neural networks to estimate unknown dynamics and to compensate for differential effects in Luenberger observers [19]. These approaches have proven successful in various applications, including the van der Pol oscillator and robotic systems.

In this paper, the mathematical models of the fiber laser, the state observer and the artificial neural network are presented in Section 2; the methodology and a description of the process are presented in Section 3; the results obtained in the implementation of the state observer and the RWFONN are presented in Section 4; the discussion is presented in Section 5; and the conclusion of the paper is presented in Section 6.

2. Mathematical Models

This section describes the three mathematical models used in the proposed implementation: the EDFL model, the state observer model, and the RWFONN model.

2.1. Erbium-Doped Fiber Laser Mathematical Model

The dynamics of a laser operating in a single emission mode can be described by a system of three coupled differential equations that consider the evolution of the optical field, the population inversion and the polarization. Each of these variables relaxes on different time scales. In certain domains, this difference allows for adiabatic approximations where one or more variables evolve instantaneously with respect to the others. These approximations simplify the mathematical model by reducing the number of equations required to describe the system's behavior.

Lasers are commonly classified into three categories according to the relative relaxation times of the optical field, polarization, and population inversion. In Class A lasers, both the polarization and population inversion relax much faster than the optical field, typically resulting in a single stable steady state. Class B lasers, where only the polarization relaxes rapidly, exhibit relaxation oscillations arising from energy exchange between the optical field and population inversion. In contrast, Class C lasers, where all three variables evolve on comparable time scales, can display complex dynamics such as sustained oscillations or deterministic chaos. Moreover, external perturbations—such as parameter modulation—may induce periodic or chaotic behavior even in Class B lasers, a phenomenon widely reported in solid-state and semiconductor devices [20–24].

From the point of view of nonlinear dynamics, fiber lasers doped with rare earths (such as erbium) are also considered Class B systems [25]. For such lasers, the rapidly relaxing polarization variable is usually eliminated by adiabatic approximation, reducing the model to two rate equations: one for the optical field and another for the population inversion. While the nonlinear behavior of lasers has been extensively studied, the analysis of erbium-doped fiber lasers (EDFLs) is a more recent endeavor. Experimental results indicate that EDFLs share the basic dynamical properties of Class B systems, including the occurrence of chaotic [26,27]. Importantly, a more comprehensive model—one that retains all three dynamical variables without adiabatic simplifications—has been developed and validated in previous studies. This complete model allows for a more accurate representation of transient phenomena, nonlinear interactions, and complex behaviors such as self-pulsation, self-modulation, and chaos induced by internal or external perturbations [28].

Normalized Equations of EDFL

In order to achieve synthesis and generalization of the laser model, the complete system was put into the following compact form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The system can be expressed in vector notation as

$$\frac{d\mathbf{x}}{d\theta} = f(\mathbf{x}), \quad (1)$$

where

$$f(\mathbf{x}) = \begin{bmatrix} xy - c_1x + c_2y + c_3 \\ -xy - d_1y - d_2 + P_0(1 - d_3e^y) \end{bmatrix}. \quad (2)$$

This can be rewritten as

$$\frac{d\mathbf{x}}{d\theta} = A\mathbf{x} + \mathbf{g}(\mathbf{x}) + \mathbf{b}, \quad (3)$$

where

$$A = \begin{bmatrix} -c_1 & c_2 \\ 0 & -d_1 \end{bmatrix}, \quad \mathbf{g}(\mathbf{x}) = \begin{bmatrix} xy \\ -xy - P_0 d_3 e^y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} c_3 \\ P_0 - d_2 \end{bmatrix}.$$

x represents the normalized laser intensity, while y corresponds to the population inversion. The parameter P_0 denotes the pump power, and P_p denotes the pump modulation. All additional parameters are given in the Table 1.

Table 1. Parameters and values of normalized equations.

Parameters and Values	Parameters and Values
$x = \frac{\sigma_{12}\Gamma_s T_r \alpha_p}{2\pi r_w^2 \alpha_0} \frac{\xi_1}{\xi_1 - \xi_2} P$	$y = \alpha_p L \left(N - \frac{1}{\xi_1} \right)$
$\theta = \frac{2r_w \alpha_0}{T_r \alpha_p} (\xi_1 - \xi_2)$	$P_0 = \frac{\alpha_p^2 T_r}{2\pi r_w^2 N_0 r_w \alpha_0 (\xi_1 - \xi_2)} P_p$
$c_1 = \frac{\alpha_p L}{\xi_1 - \xi_2} \left(\frac{\alpha_{th}}{\alpha_0 r_w} + \frac{\xi_2}{\xi_1} \right)$	$c_2 = \frac{\xi_1 \alpha_p}{\pi r_w \alpha_0} \frac{T_r}{\tau} \left[\frac{\lambda_g}{4\pi w_0 (\xi_1 - \xi_2)} \right]^2 \times 10^{-3}$
$c_3 = \frac{L}{\pi r_w \alpha_0} \frac{T_r}{\tau} \left[\frac{\lambda_g \alpha_p}{4\pi w_0 (\xi_1 - \xi_2)} \right]^2 \times 10^{-3}$	$d_1 = \frac{\alpha_p}{2r_w \alpha_0 (\xi_1 - \xi_2)} \frac{T_r}{\tau}$
$d_2 = \frac{\alpha_p^2 L}{2r_w \alpha_0 \xi_1 (\xi_1 - \xi_2)} \frac{T_r}{\tau}$	$d_3 = \exp \left[-\alpha_p L \left(1 - \frac{1}{\xi_1} \right) \right]$

The values of the parameters used in the construction of the state observer are listed in Table 2 [29].

Table 2. Parameters for construction of state observer [30,31].

c_1	c_2	c_3	d_1	d_2	d_3	P_0
2.4	6.9×10^{-13}	5.1×10^{-13}	3.5×10^{-7}	2.6×10^{-7}	0.5	$2 \times 10^{-23} P_p$

2.2. Mathematical Model of State Observer

In dynamic systems, especially nonlinear systems, direct access to all state variables is often not possible due to physical, technological or cost limitations of instrumentation. Given this limitation, nonlinear state observers are indispensable tools for real-time estimation of non-measurable variables. To construct the state observer, we define the state vector and the input, using as reference Equation (3):

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad u = u_{\text{pump}}$$

Since the system contains nonlinear terms such as x, y , it is generally represented as a nonlinear system:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} \quad (4)$$

where

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} xy - c_1 x + c_2 y + c_3 \\ -xy - d_1 y - d_2 \end{bmatrix}, \quad \mathbf{g}(\mathbf{x}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (5)$$

So the complete system is expressed as

$$\dot{\mathbf{x}} = \begin{bmatrix} xy - c_1 x + c_2 y + c_3 \\ -xy - d_1 y - d_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_{\text{pump}} \quad (6)$$

The output of the system is defined as

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x} \quad (7)$$

For the construction of the state observer, we define the vectors

$$\tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix}, \quad u = u_{\text{pump}}$$

where L_1 and L_2 are gains of the observer. Then, the nonlinear observer can be expressed as

$$\dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \tilde{x}\tilde{y} - c_1\tilde{x} + c_2\tilde{y} + c_3 \\ -\tilde{x}\tilde{y} - d_1\tilde{y} - d_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} (y - \tilde{y}) \quad (8)$$

where $x_1 - \tilde{x}_2 = e_{x0}$ is named the estimated error. The estimated output is

$$\tilde{y} = \begin{bmatrix} 1 & 0 \end{bmatrix} \tilde{\mathbf{x}} \quad (9)$$

In this paper, a state observer is developed to estimate the state variable corresponding to population inversion (y). This decision is made due to the impossibility of measuring this variable directly in real time and the lack of suitable electronic instruments for its collection. The stability analysis of the proposed observer is presented in Appendix A.

2.3. Mathematical Model of RWFONN

In the present work, a Recurrent Wavelet First-Order Neural Network (RWFONN) was chosen due to its advantageous properties over other neural network architectures commonly used in the identification of nonlinear systems. One of the main reasons for this choice lies in its ability to estimate the internal states of the dynamical system in real time by using an adaptive mechanism that updates its synaptic weights online based on the filtered estimation error. This feature makes the RWFONN particularly suitable for observer-based architectures in systems where certain internal variables, such as population inversion in EDFL, cannot be measured directly. In addition, its wavelet-based structure enables a compact representation of the signal with strong time–frequency localization, which improves the generalizability and convergence speed of the network. Compared to more complex models such as the Recurrent High-Order Neural Network (RHONN) or the Recurrent Sigmoid First-Order Neural Network (RSFONN), the RWFONN has a simpler architecture with lower computational complexity, making it more practical for real-time control applications and environments with limited hardware.

In [32], the authors identify a dynamical UDS type I by numerical approximation using an RWFONN, where the general structure is as follows:

$$\dot{y}_j^i = -\alpha_j^i y_j^i + (w_{jk}^i)^\top \psi_{jk}^i, \quad (10)$$

where y_j^i are the states of the i th neuron. $\alpha_j^i > 0$ for $i = 1, 2, \dots, n$ is part of the underlying network architecture and remains fixed during the training process. w_{jk}^i is the k th adjustable synaptic weight connecting the j th state to the i th neuron, and ψ_{jk}^i is a Morlet wavelet activation function, which is defined by $\psi(\chi) = e^{(-\chi^2/\beta)} \cos(\mu\chi)$, where χ is the state of the original system to be identified; the parameters β and μ are the expansion and dilation terms. The system given in (3) is identified online using the RWFONN, where the synaptic weights are adjusted using the filtered error algorithm. A more detailed description of the network structure can be found in [13,28].

3. Methodology and Description of the Process

3.1. Experimental Setup

Figure 1 shows the experimental configuration of the erbium-doped fiber laser (EDFL). The cavity, with a total length of 6.5 m, includes a 70 cm segment of erbium-doped fiber (EDF) (SM-EDF-7/125, Thorlabs, Newton, NJ, USA) with a core diameter of $2.7\ \mu\text{m}$. The EDF is optically pumped by a 977 nm laser diode (LD-BL976PAG500, Thorlabs, Newton, NJ, USA), whose current (black arrows means electric current, while red arrows mean optical signals) is regulated by a Laser Diode Controller (LDC-ITC510, Thorlabs, Newton, NJ, USA) that also ensures thermal stabilization of the pump source. The pump current is modulated using a sinusoidal signal $m \sin(2\pi f_0 t)$ generated by a waveform generator (WFG-AFG3102, Tektronix, Newton, NJ, USA), which is connected to the modulation input of the LDC. The optical signal from the LD passes through a Polarization Controller (PC) (FPC020, Thorlabs, Newton, NJ, USA) and is injected into the cavity via a Wavelength Division Multiplexer (WDM-WD9850FD, Tektronix, Beaverton, OR, USA). The resonator comprises two Fiber Bragg Gratings (FBG1 and FBG2, Thorlabs, Newton, NJ, USA) centered at 1550 nm, with reflectivities of 100% and 95.88%, respectively. The cavity also includes an Optical Isolator (IO-H-1550APC, Thorlabs, Newton, NJ, USA) at the FBG2 port to eliminate back-reflections.

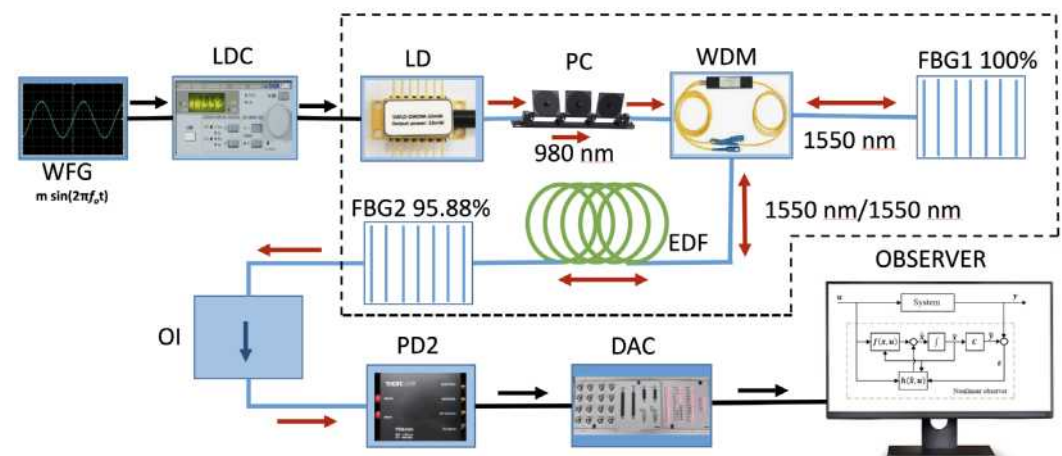


Figure 1. The experimental setup of the EDFL system. The output intensity is acquired and processed by a nonlinear observer that estimates the state of the population inversion. The observer compares experimental and numerical intensity for real-time analysis.

The output laser signal from the FBG2 port is detected by a photodiode (PD2, PDA10CS-AC, Thorlabs, Newton, NJ, USA), and the resulting electrical signal is digitized through a data acquisition card DAC (DS1104, dSPACE, Paderborn, DE). The sampled signal is processed by a nonlinear state observer (see Section 2.2), as explained in the following Section 3.2, which estimates non-measurable internal states such as population inversion. This observer also enables the comparison between experimental and numerically reconstructed signals and can optionally provide feedback to the modulation input via the WFG-LDC interface. All optical components are based on a single-mode fiber (SMF-28, Thorlabs, Newton, NJ, USA) with a cladding diameter of $200\ \mu\text{m}$. During all measurements, the EDFL system was kept under active thermal stabilization.

3.2. Schematic Implementation of State Observer in Real-Time

Figure 2 shows a real-time schematic of the implementation of the nonlinear state observer applied to the EDFL system. The diagram consists of three main processes. The first block, labeled EDFL experimental system,

represents the physical laser setup used to measure the laser intensity x and send it to the observer. In the second block, labeled state observer, a nonlinear observer both estimates the measured variable $\tilde{x}$ and reconstructs the unmeasured internal state $\tilde{y}$, which corresponds to population inversion. The dynamics of the observer are determined by a system of differential equations, which are shown in the block. The error between the measured and estimated outputs controls the estimation.

These estimated states $(\tilde{x}, \tilde{y})$ are then entered into the third block, RWFONN, which performs the system identification by generating the outputs y_1 and y_2 based on the neural model. In addition, the diagram contains the error signals e_{xo} , e_{yor} , and e_{xor} , which quantify the estimation and identification accuracy by comparing observed, estimated, and neural network outputs.

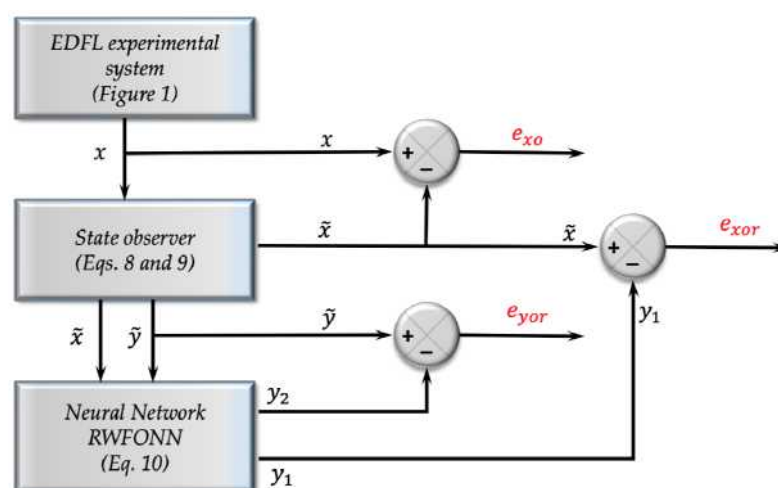


Figure 2. A schematic of the implementation of the nonlinear observer for the EDFL system. The observer estimates laser intensity and reconstructs population inversion in real time. A neural network (RWFONN) validates the observer’s performance using the estimated states.

3.3. Temporary Rescaling

Since the integration step size for the numerical solution of the differential equations describing the dynamics of the observer in Simulink is very small compared to the sampling rate of the real-time DAC, the observer equation is temporarily rescaled as follows: If we consider $t = H\tau$ with $st = Hd\tau$, where H is the scaling parameter of the observer, the equation changes to

$$\frac{dx}{Hd\tau} = \tilde{x}\tilde{y} - c_1\tilde{x} + c_2\tilde{y} + c_3 + L_1(x - \tilde{x}) \quad (11)$$

$$\frac{dy}{Hd\tau} = -\tilde{x}\tilde{y} - d_1\tilde{y} - d_2 + u_{pump} + L_2(x - \tilde{x}) \quad (12)$$

and by multiplying Equations (11) and (12) by H , these become

$$\frac{dx}{d\tau} = H(\tilde{x}\tilde{y} - c_1\tilde{x} + c_2\tilde{y} + c_3 + L_1(x - \tilde{x})) \quad (13)$$

$$\frac{dy}{d\tau} = H(-\tilde{x}\tilde{y} - d_1\tilde{y} - d_2 + u_{pump} + L_2(x - \tilde{x})) \quad (14)$$

If you set the parameter H to a suitable value, the observer model implemented in Simulink and the real-time acquisition platform can work with the DAC in real time.

4. Real-Time Observer and Neural Identification Results

To evaluate the performance and practicality of the proposed nonlinear state observer, the system was implemented in real time using the RWFONN for state reconstruction. In this section, the experimental results obtained with an erbium-doped fiber laser, where direct measurement of critical internal states, such as population inversion, is not possible, are presented. The observer was designed to estimate these unmeasurable variables from the measured laser intensity, and the RWFONN was trained online with the filtered estimation error to adaptively improve the accuracy of the reconstruction. The following results demonstrate the effectiveness of the proposed architecture under different operating conditions and show its convergence behavior, its robustness to unmodeled dynamics, and its suitability for real-time control scenarios.

Using the parameters of Table 3 and the structure of the artificial neural network to identify the state variables obtained by the state observer is as follows:

The state vector

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},$$

the system matrix, which contains the linear coefficients of the variables y_1 and y_2 , is defined as

$$\mathbf{A} = \begin{bmatrix} -a_1 & 1 \\ 0 & -a_2 \end{bmatrix}$$

The vector of wavelet activation functions evaluated is

$$\boldsymbol{\psi}(\boldsymbol{\chi}) = \begin{bmatrix} \psi(\chi_1) \\ \psi(\chi_2) \end{bmatrix}$$

The weighted parameters for each nonlinear input are grouped into the following vector:

$$\mathbf{b} = \begin{bmatrix} b_1 w_1 \\ b_2 w_2 \end{bmatrix}$$

Finally, the external input to the system is represented as

$$\mathbf{u} = \begin{bmatrix} 0 \\ u_{\text{pump}} \end{bmatrix}$$

Then, the equations can be written in matrix form as

$$\dot{\mathbf{y}} = \mathbf{A}\mathbf{y} + \mathbf{b} \circ \boldsymbol{\psi}(\boldsymbol{\chi}) + \mathbf{u} \quad (15)$$

where $\circ$ represents the element-by-element product.

Table 3. Parameters of the state observer and RWFONN.

Parameters	Parameters
$L_1 = 4.083 \times 10^8$	$L_2 = 3.5 \times 10^4$
$H = 1 \times 10^{-5}$	$a_1 = a_2 = 2460$
$b_1 = b_2 = 2460$	$\lambda_1 = \lambda_2 = 1 \times 10^{-6}$
$\beta_1 = 9.5 \times 10^4$	$\beta_2 = 22.5 \times 10^4$
$\gamma_1 = 3123$	$\gamma_2 = 3123$

Figure 3 shows the laser intensity and the reconstructed population inversion of an EDFL under three different initial conditions in each row. These conditions are from independent experimental acquisitions at 80 kHz, where the measurement system, consisting of a photodetec-

tor, DAC and computer, was reinitialized before each run. This variability tests the robustness of the proposed estimation system. The left column Figure 3a–c shows the laser intensity signals: experimental measurements (red dashed line), estimates from the nonlinear state observer (blue solid line) and predictions by the RWFONN (black dashed line). The right column Figure 3d–f shows the corresponding reconstructions of the non-measurable state variable, the population inversion, from both the observer (red dashed line) and the neural network (black solid line). The consistent agreement between the observer and the neural network across all three initial conditions confirms the effectiveness and repeatability of the hybrid approach for real-time state reconstruction in complex EDFL systems.

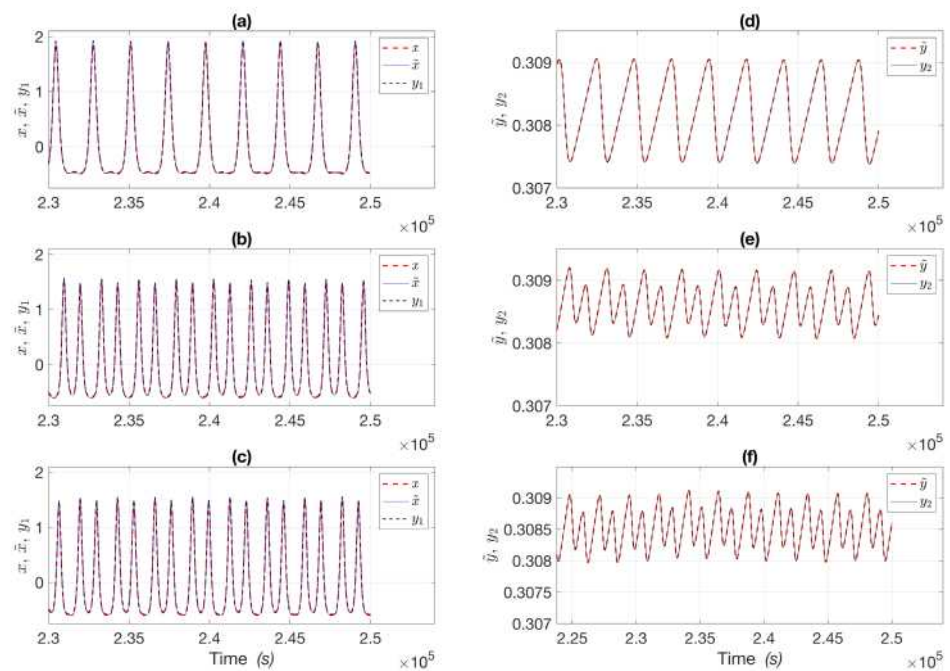


Figure 3. Laser intensity and population inversion estimations for an EDFL. The left column (a–c) shows the experimental laser intensity (red dashed line), observer estimate (blue continuous line) and RWFONN output (black dashed line). The right column (d–f) shows the population inversion estimated by the observer (red dashed line) and RWFONN (black continuous line). Each row corresponds to different initial conditions from the acquisition system.

Figure 4 shows the results of the real-time observer and identification of the laser intensity. The results shown in the graphs were obtained for different frequency values. For this study, the frequencies of interest were considered to be every 10 kHz. The efficiency of the observer and the artificial neural network is indicated in achieving both estimation and identification. Each figure shows a box with detail, in which good estimation and identification of laser intensity can be observed.

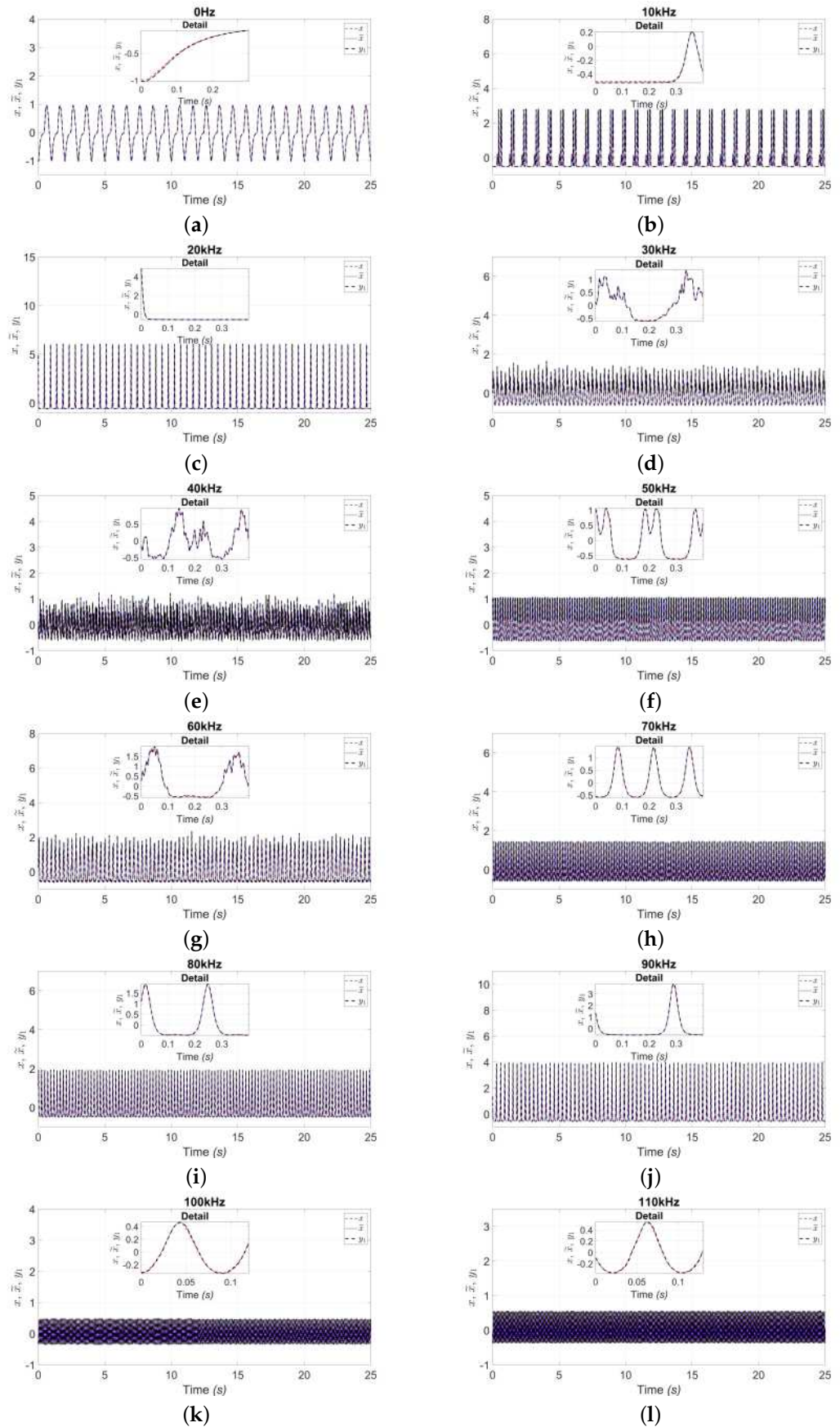


Figure 4. Cont.

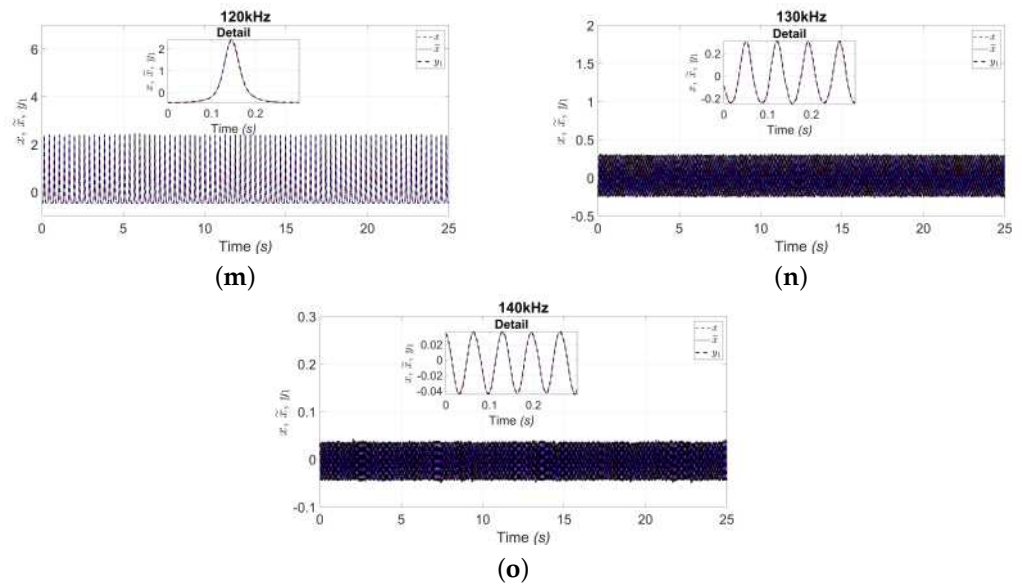


Figure 4. Real laser intensity (red dashed line), state observer (blue solid line) and artificial neural network (blue solid line), respectively, for every value of frequency, (a) 0 Hz, (b) 10 kHz, (c) 20 kHz, (d) 30 kHz, (e) 40 kHz, (f) 50 kHz, (g) 60 kHz, (h) 70 kHz, (i) 80 kHz, (j) 90 kHz, (k) 100 kHz, (l) 110 kHz, (m) 120 kHz, (n) 130 kHz and (o) 140 kHz.

Figure 5 shows the results of the estimated population inversion and neuronal identification. As with the laser intensity plots, the results shown in the graphs were obtained for different frequency values. The frequencies of interest were all considered 10 kHz. The efficiency of the observer and the artificial neural network can be seen in both estimation and identification. Each figure shows a box with a detail where good estimation and identification of the population inversion can be observed.

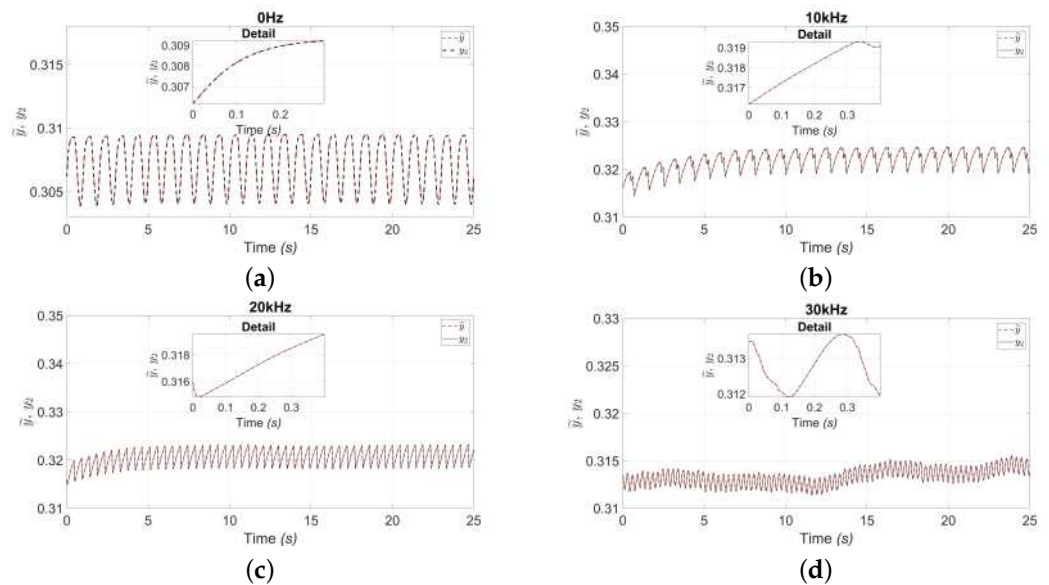


Figure 5. Cont.

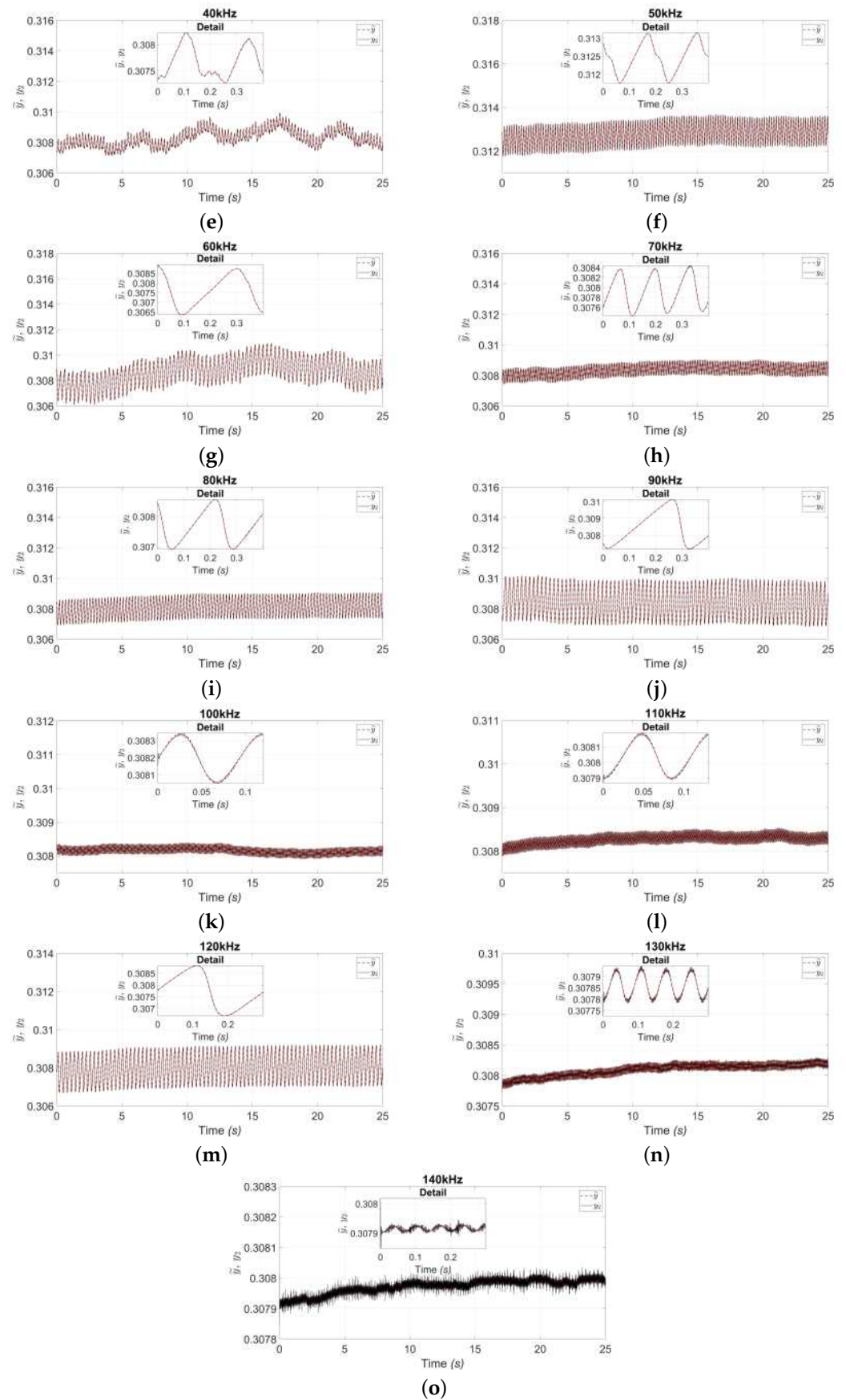


Figure 5. Estimated population inversion (red dashed line) and artificial neural network output (black solid line), respectively, for every value of frequency: (a) 0 Hz, (b) 10 kHz, (c) 20 kHz, (d) 30 kHz, (e) 40 kHz, (f) 50 kHz, (g) 60 kHz, (h) 70 kHz, (i) 80 kHz, (j) 90 kHz, (k) 100 kHz, (l) 110 kHz, (m) 120 kHz, (n) 130 kHz and (o) 140 kHz.

Euclidean Distance and MSE Metrics

Two metrics were used to quantitatively validate the accuracy of the proposed state observer and the implemented artificial neural network: Euclidean distance [33] and mean square error (MSE). These metrics allow for comparison of the behavior of observed and estimated variables with respect to simulated or measured references, and provide an objective criterion to evaluate the system performance. Specifically, the Euclidean distance between the trajectories of the estimated and actual variables was calculated, and the MSE was used to measure the degree of error dispersion over time. These analytical tools were applied to different pairs of relevant system variables, allowing the degree of agreement and reliability of the proposed model to be determined when estimating variables that could not be measured directly. The Figure 6 shows the measure of Euclidean distance metric at 80 kHz of frequency.

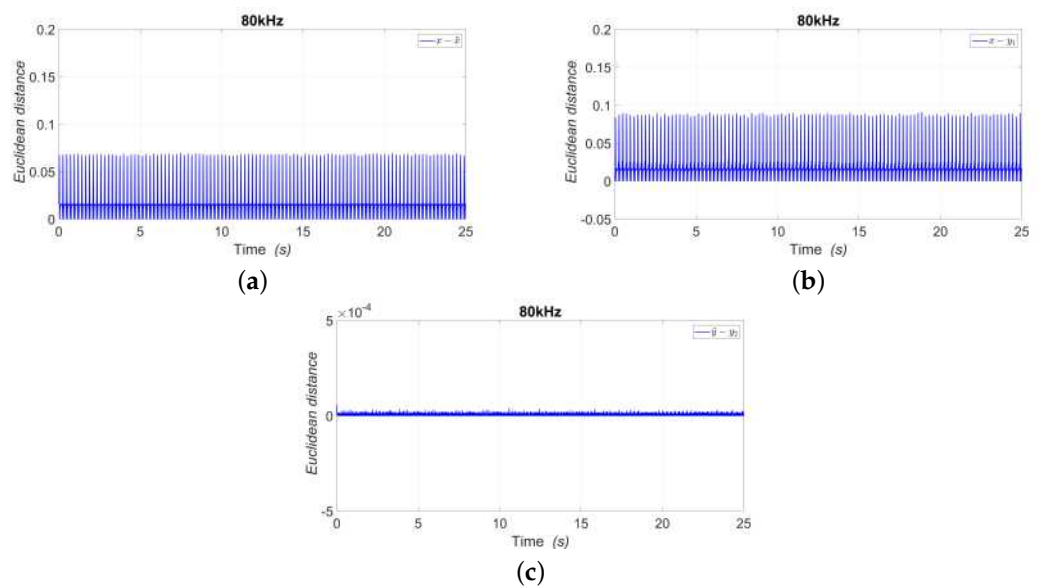


Figure 6. Euclidean distance metric to 80 kHz value of frequency. Difference between (a) laser intensity and first variable estimated by state observer ($x - \hat{x}$), (b) laser intensity and first variable of RWFONN ($x - y_1$), and (c) second variable estimated by state observer and second variable of RWFONN ($\tilde{y} - y_2$).

To verify the accuracy of the estimates under steady-state conditions, the selected metrics were applied to the case where the input signal to the system was 80 kHz. In this scenario, three specific combinations were compared: the difference between the real laser intensity and the observed state ($x - \hat{x}$), the difference between the real laser intensity and the first variable of the neural network ($x - y_1$), and finally the difference between the variable estimated by the state observer and the second variable of the neural network ($\tilde{y} - y_2$). In all three scenarios, it can be observed that the error is close to zero, which guarantees good observation and identification of the laser system.

The results obtained in Table 4, which calculates the mean square error for each evaluated frequency value, consistently show values close to zero. This trend indicates high accuracy of the proposed state observer, since the difference between the estimated and the actual signals is minimal throughout the operating range. Consequently, the correct reconstruction of the non-measurable variables and adequate identification of the dynamic behavior of the system are confirmed. These results support the reliability of the approach chosen for real-time estimation within the EDFL.

Table 4. MSE for every value of frequency.

Frequency	$(x - \tilde{x}) \times 10^{-4}$	$(x - y_1) \times 10^{-4}$	$(\tilde{y} - y_2)$
10 kHz	10.0	20.0	1.008
20 kHz	30.0	50.0	2.449
30 kHz	3.91	7.71	0.412
40 kHz	3.25	10.0	0.265
50 kHz	5.23	9.51	0.521
60 kHz	7.11	10.0	0.701
70 kHz	5.81	10.0	0.565
80 kHz	6.31	9.33	0.652
90 kHz	10.0	20.0	1.426
100 kHz	1.06	2.13	0.177
110 kHz	1.42	3.10	0.200
120 kHz	7.05	10.0	0.725
130 kHz	0.56	1.36	0.135
140 kHz	0.01	0.03	0.097

5. Discussion

The implementation of the proposed state observer demonstrates the feasibility of estimating the population inversion in an erbium-doped laser fiber. This variable cannot be measured directly due to the physical limitations of the system. The estimation is based on measurement of the laser output intensity, which allows the overall dynamics of the system to be derived with an acceptable margin of error. This strategy provides an efficient alternative for monitoring internal variables in complex optical systems without changing their physical architecture or interrupting their operation.

The inclusion of an artificial neural network enabled the validation of the behavior estimated by the observer and showed a significant correlation between the two methods. In addition, the ANN proved capable of capturing the nonlinearities inherent in the system, indicating its usefulness as a backup or support under more demanding operating conditions or when the physical parameters of the laser fluctuate.

An important observation is that while the state observer offers a more transparent solution from a physical and mathematical point of view, the neural network offers flexibility and generalization possibilities if sufficient training data is available. However, its performance strongly depends on the quality of the data and the chosen architecture.

Overall, the combination of both methods not only improves the robustness of the estimation system, but also opens up the possibility of developing more advanced hybrid strategies for the control and monitoring of laser systems. This research lays the foundation for future real-time implementations in more complex experimental setups.

6. Conclusions

The results obtained in this work demonstrate the effectiveness of the proposed state observer in conjunction with the RWFONN for the estimation and reconstruction of the internal variables in an erbium-doped laser system. In particular, the second state variable, which cannot be measured directly due to the physical limitations of the system, was estimated in real time. Moreover, both observed state variables could be accurately identified, confirming the ability of the proposed scheme to robustly and adaptively represent the system dynamics. This approach not only reduces the need for invasive or expensive instrumentation but also lays the foundation for the development of future advanced control systems for nonlinear optical systems. The RWFONN structure, with its online learning and low computational cost, proves to be particularly well-suited for real-time applications where efficiency and accuracy are critical.

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Appendix A. Stability Analysis

Analyzing the stability of the error system between the original laser (Equation (6)) and the state observer (Equation (8)), the representation in terms of the dynamic estimated error is defined as follows:

$$\begin{aligned} \dot{e}_1 &= xy - c_1x + c_2y + c_3 - [\tilde{x}\tilde{y} - c_1\tilde{x} + c_2\tilde{y} + c_3 + L_1(x - \tilde{x})], \\ \dot{e}_2 &= -xy - d_1y - d_2 + P_{pump} - [-\tilde{x}\tilde{y} - d_1\tilde{y} - d_2 + P_{pump} + L_2(x - \tilde{x})]. \end{aligned} \quad (\text{A1})$$

Reducing and factoring, Equation (A1) can be rewritten as follows:

$$\begin{aligned} \dot{e}_1 &= (\tilde{y} - c_1 - L_1)e_1 + (x + c_2)e_2, \\ \dot{e}_2 &= -(\tilde{y} + L_2)e_1 - (x + d_1)e_2. \end{aligned} \quad (\text{A2})$$

The next positive definite Lyapunov function guarantees that the error tends to zero:

$$V(e) = \frac{1}{2}(e_1^2 + e_2^2), \quad (\text{A3})$$

Thus, the derivative of the Lyapunov function becomes

$$\dot{V}(e) = (\tilde{y} - c_1 - L_1)e_1^2 + (x + c_2 - \tilde{y} + L_2)e_1e_2 - (x + d_1)e_2^2. \quad (\text{A4})$$

On the other hand, Equation (A4) can be written in matrix form to find the values of L_1 and L_2 , in order to find the optimal gains that guarantee the stability of the whole system:

$$\dot{V}(e) = - \begin{bmatrix} e_1 & e_2 \end{bmatrix} \begin{bmatrix} c_1 + L_1 - \tilde{y} & -\frac{1}{2}(x + c_2 + L_2 - \tilde{y}) \\ -\frac{1}{2}(x + c_2 + L_2 - \tilde{y}) & x + d_1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \quad (\text{A5})$$

in the form $e^T P e$, with $\dot{V} < 0$, if

$$L_1 > 0;$$

and

$$L_2 < \sqrt{(L_1 - \tilde{y} + c_1)(x + d_1)} - (x + c_2 - \tilde{y}). \quad (\text{A6})$$

With the real measured values $|x|_{\max} \approx 6$, $|\tilde{y}|_{\max} \approx 0.3$ and the constants given in Table 2, the choice of $L_1 = 4083 \times 10^5$ leads to $L_2 \lesssim 49,490$. Then $L_2 = 35,000$ fulfills the necessary condition (A6). This guarantees the asymptotic stability of the entire system.

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